HackResearch 2026

Decoding Cellular Switches

Biological background

Kinases add phosphate groups to specific positions on proteins; phosphatases remove them. This reversible modification, called phosphorylation, can alter a protein's activity, location or interactions. The protein being modified is a substrate, and the position is a phosphosite: A-S10 means serine at position 10 of protein A. Understanding which enzymes act on which sites helps explain how cells regulate their behavior. In our datasets, kinase targets are much better documented than phosphatase targets.

The research problem

Which proteins and phosphorylation sites are targeted by each phosphatase? This project asks whether richer kinase–substrate data, combined with biological context, can reveal missing phosphatase–substrate associations through heterogeneous graphs or statistical inference.

The central task is to predict plausible missing links between phosphatases and their substrate proteins. Participants will determine how to represent and combine the supplied data, develop a prediction approach, and assess the evidence supporting its predictions. Which information is useful for this task? What makes a proposed relationship biologically convincing? These are open research questions. An unrecorded relationship may be undiscovered rather than biologically absent, and incomplete or uneven data coverage is part of the problem.

Available data

One shared folder will contain kinase–substrate associations (KSA from PhosphositePlus), phosphatase–substrate associations (PSA, from DEPOD), MSigDB pathway and gene-set memberships, drug–disease associations, drug–target associations, and protein–protein interactions (PPI). Together, these describe known enzyme targets and their molecular and disease context.

Data folder link: Data

Potential contribution

Any identified pattern, an interpretable model or statistical analysis that identifies candidate phosphatase targets and establishes when kinase and contextual information add value beyond simple connectivity. Results should support testable biological hypotheses and make their uncertainty clear. The challenge welcomes graph algorithms, machine learning and statistical thinking.